

Internship proposal

Optimal recovery of joint probabilities using cluster graphs

The CIPHOD team at Sorbonne Public Health Institute is seeking an M2 intern for spring 2027 who is motivated by graphical approaches to causal inference and missing data problems.

Context: Missing data is a major challenge in public health, where incomplete follow-up, non-response, or unavailable measurements can lead to biased conclusions. Causal graphical models provide a principled framework for reasoning about missingness and determining when probabilistic or causal quantities can be recovered from incomplete data Hernan and Robins [2023], Pearl [2009], Mohan et al. [2013], Mohan and Pearl [2014, 2021]. However, specifying a complete variable-level directed acyclic graph (DAG) is often unrealistic. In many applications, domain experts may only be able to provide causal information between groups of variables rather than between every individual variable. Cluster graphs offer a way to represent such partial causal knowledge. Recent work has studied causal identification from cluster-level graphs Ferreira and Assaad [2025], and recent results provide conditions for recovering joint distributions from cluster graphs in the presence of missing data Scott et al. [2026]. These results establish whether a target distribution is recoverable, but they do not necessarily determine which recovery formula should be preferred when several valid formulas exist.

Proposal: The objective of this internship is to investigate the statistical efficiency of different recovery formulas implied by the same cluster graph. Suppose several valid recovery formulas identify the same target distribution:

$$P(\mathbf{V}) = \Phi_1(P_{\text{obs}}) = \dots = \Phi_K(P_{\text{obs}}).$$

Although these formulas recover the same population quantity, their empirical estimators may have different finite-sample and asymptotic variances.

The first goal of the internship will be to investigate this question empirically. Through simulation studies, we will compare different valid recovery formulas in terms of bias, variance, and mean squared error, and study how their performance depends on the underlying graph and missingness mechanism.

Based on the patterns observed empirically, the second goal will be to characterize when one recovery formula is more efficient than another, and to investigate whether properties of the cluster graph can be used to identify an optimal or low-variance recovery strategy.

Candidate: Highly motivated candidate with an M2 or engineering degree and strong background in probability, machine learning, and causal inference, along with an interest in epidemiology. Proficiency in programming is also required. Please note that we will not consider any M1 applicants for this position.

Location: The intern will work at Sorbonne Public Health Institute (<https://iplesp.fr/>), located in Paris. They will be supervised by Willow Scott and Charles Assaad.

Dates: Starting date: To be discussed, early 2027, for a duration of 5-6 months.

Contact: To apply, please send a CV and a cover letter to Willow Scott willow.scott@inserm.fr

References

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- M. Hernan and J. Robins. *Causal Inference: What If*. Chapman & Hall/CRC Monographs on Statistics & Applied Probab. CRC Press, 2023. ISBN 9781420076165.
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- J. Pearl. *Causality*. Cambridge university press, 2009.
- W. Scott, E. Valdano, and C. K. Assaad. Missing data and cluster graphs: cluster-level missingness vs variable-level missingness, 2026.